

Preface

Numerical Analysis

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Chapter 1

Finite Difference Approximations

§ 1.1. Preliminaries

Definition 1.1.1. Let f and g be real or complex-valued functions on a domain D . We say that f is *big O of* g if there exists positive real numbers M and δ such that for all $h \in D$, if $0 < |h| < \delta$ then $\left| \frac{f(h)}{g(h)} \right| < M$.

Example 1.1.1. We have that $2h^3 = O(h^3)$ as $h \rightarrow 0$. Indeed, we can see:

$$\left| \frac{2h^3}{h^3} \right| = 2$$

for any arbitrarily small δ .

for any sufficiently small h . Now consider $\delta = \frac{1}{2}$.

$$\left| \frac{2h^3}{h^2} \right| = 2|h| \leq 1,$$

hence